

YASHASWINI R

AI Engineer | Bangalore, India

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PROFESSIONAL SUMMARY

AI Engineer who builds and ships enterprise-grade Generative AI systems — knowledge graphs, hybrid retrieval, and agentic RAG pipelines — end to end. Built a governed enterprise Semantic Layer on Neo4j that scaled to 1M+ data points at 5–10s p95 latency, and engineered hybrid retrieval (vector + full-text + graph, RRF-fused, cross-encoder reranked) evaluated against RAGAS quality gates. Outside of work, independently designs and ships full AI products solo — from user research to a deployed, multilingual, agentic prototype — using LangChain, LangGraph, FastAPI, and MCP.

EXPERIENCE

Neftura Renewables Pvt. Ltd., Bangalore

AI Engineer

May 2026 – Present

- Built a governed enterprise Semantic Layer, transforming structured and unstructured enterprise data into a Neo4j knowledge graph that scaled to 1M+ data points while maintaining 5–10s p95 query latency.
- Developed structured and unstructured document ingestion pipelines — schema inference, chunking, and LLM-based entity/relationship extraction — using Neo4j and LangChain's LLMGraphTransformer, with incremental hash-based reprocessing to avoid full re-ingestion on updates.
- Implemented hybrid retrieval combining vector search, full-text search, and graph traversal, fused via Reciprocal Rank Fusion (RRF) and reranked with a SentenceTransformer cross-encoder — contributing to Precision@K ~60% and Recall@K ~70% against target metrics.
- Contributed to evaluating RAG system quality using RAGAS metrics (Faithfulness, Precision@K, Recall@K) against stage-wise quality gates spanning ingestion through generation, supporting a system-wide Faithfulness score ≥ 0.85 .

AI Engineer Intern

Dec 2025 – May 2026

- Built an end-to-end RAG pipeline — document embedding, retrieval, and LLM-based response generation — that became the foundation for the hybrid retrieval and evaluation systems later developed in the full-time role.

PROJECTS

Agentic AI Career Launch Agent

github.com/Yashu-ram/agent-ai-career-launch-agent

FastAPI · ChromaDB · Sentence Transformers · Ollama + Qwen2.5 · LangChain · LangGraph · RAG · Pydantic

- Built a RAG-based career guidance assistant analyzing resumes against job descriptions via ChromaDB and Sentence Transformers, running on a locally-hosted LLM (Ollama + Qwen2.5) rather than a paid hosted API.
- Implemented a LangGraph-powered multi-agent pipeline for structured career guidance and missing-skill identification, achieving 89% skill-match accuracy (8 of 9 test cases) with all structured outputs validated against Pydantic schemas at zero error (9/9).
- Integrated AI safety guardrails and prompt-engineering strategies to reduce hallucination risk in the pipeline, validated through manual review across 9 edge-case scenarios.

RTI Mitra – AI-Assisted RTI Filing Prototype

rti-mitra.vercel.app

UX Research · AI Product Design · Codex-Assisted Prototyping · Graph RAG · Vercel

- Researched pain points in India's RTI Online process — including authority discovery, confusing forms, payment failures, tracking delays, and first-appeal difficulty — through independent UX research to define the product scope.
- Built Graph RAG-style authority routing over a 2,916-entry public authority directory (RTI Online), enabling citizens to identify the correct authority from plain-language queries.
- Designed and shipped a Vercel-deployed prototype covering RTI draft generation, status tracking, and first-appeal assistance, with voice input and privacy checks — validated end-to-end against a mock backend, supporting 15+ languages.
- Architected the system with a decoupled frontend (Vercel), a serverless API layer, and a mock backend service — structuring the Graph RAG authority-routing and draft-generation logic as independent, swappable modules for future integration with the real RTI Online backend.

EDUCATION

Master of Science — St. Joseph University, Bangalore

2023 – 2025

Bachelor of Science — Bangalore City University, Bangalore

2020 – 2022

CERTIFICATION

Generative AI Intensive Training — Outskills Workshop

Jan 2026

Covered AI tools, LLM-based chatbots, automation workflows, no-code AI product development, AI image and video generation, and agentic AI systems through hands-on real use-case sessions.

SKILLS

AI Engineering: Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Hybrid Retrieval, Reciprocal Rank Fusion (RRF), Reranking, Semantic Search, Knowledge Graphs, LangChain, LangGraph, Prompt Engineering

Agentic AI: LangGraph (Plan → Execute → Synthesis → Reflection), Model Context Protocol (MCP)

Evaluation & Quality: RAGAS (Faithfulness, Precision@K, Recall@K), Stage-wise Quality Gates, Human-in-the-Loop Review

Knowledge Graph: Neo4j, Cypher, Entity Canonicalization, Graph Traversal, KNN Similarity Edges (Neo4j GDS)

Programming & Backend: Python, FastAPI, Pydantic, REST APIs, API Integration, Object-Oriented Programming, Git

Databases: SQL, Neo4j (Graph + Vector), Vector Databases (ChromaDB)

LLMs & AI Applications: OpenAI API, Claude, Ollama (local LLM deployment), Open-Source LLMs, AI Agents, AI Safety & Guardrails